

E ADDITIONAL TAMP SUITE DETAILS

We evaluate our framework on three task domains (`hook_reach`, `rearrangement_push`, and `rearrangement_memory`) with two tasks each. Each of the considered suites focuses on understanding long-horizon success of one particular skill. For example, `hook_reach` is about the long-term effect of executing `hook`, while `rearrangement_push` focuses on `push` and `rearrangement_memory` is designed to confuse the TAMP framework that perform hierarchical planning with non goal-conditioned motion planners. Each task’s challenge is directly proportional to the long-horizon action dependency required to complete it. For example, `pull` affects immediately if the next skill is `pick`. But `place` affects the next skill after executing one intermediate skill (like `pick`). Similarly, action dependency is after two skills for `rearrangement_push` and `rearrangement_memory` tasks. We describe all of such considered tasks below.

1. Hook Reach (Task 1):

- **Scene:** Table with a rack, hook, and cube
- **Start:** Rack and hook are in workspace, cube is beyond workspace
- **Goal:** Pick up the cube
- **Action Skeleton:** `pick(hook) → pull(cube, hook) → place(hook) → pick(cube)`

2. Hook Reach (Task 2):

- **Scene:** Table with a rack, hook, and cube
- **Start:** Rack and hook are in workspace, cube is beyond workspace
- **Goal:** Place the cube on the rack
- **Action Skeleton:** `pick(hook) → pull(cube, hook) → place(hook) → pick(cube) → place(cube, rack)`

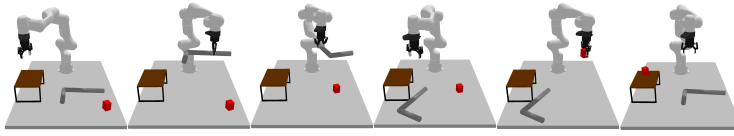


Figure 10: Hook Reach Task 2

3. Rearrangement Push (Task 1):

- **Scene:** Table with a hook, cube, and rack
- **Start:** Hook and cube are in workspace, rack is beyond workspace
- **Goal:** Position the cube under the rack
- **Action Skeleton:** `pick(cube) → place(cube) → pick(hook) → push(cube, hook, rack)`

4. Rearrangement Push (Task 2):

- **Scene:** Table with a hook, cube, and rack
- **Start:** Hook is in workspace, cube and rack are beyond workspace
- **Goal:** Position the cube under the rack
- **Action Skeleton:** `pick(hook) → pull(cube, hook) → place(hook) → pick(cube) → place(cube) → pick(hook) → push(cube, hook, rack)`

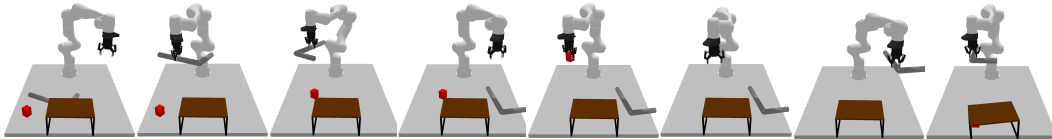


Figure 11: Rearrangement Push Task 2

5. Rearrangement Memory (Task 1):